Customer Basket Analysis - SQL Project

Project Overview

The goal is to simulate how actual data analysts in the e-commerce or retail industries work behind the scenes to use SQL to:

✅ **Set up a real-world customer transaction database**

In real-world scenarios, analysts first structure messy customer and transaction data into SQL by handling inconsistencies like missing values, age errors, and date format issues to make it analysis-ready.

✅ **Perform Exploratory Data Analysis (EDA)** to understand customer demographics, purchase patterns, and sales trends

Analysts use SQL for exploratory data analysis to uncover patterns, anomalies, and data quality issues like inactive customers, pricing errors, and purchase gaps before deeper insights can be drawn.

✅ **Implement Data Cleaning** to handle null values, remove invalid entries, and standardize date formats

Analysts clean the data using SQL by correcting invalid entries (like ages > 100) and standardizing values like dates, ensuring consistency and reliability for accurate business analysis.

✅ **Write business-driven SQL queries** to derive insights around **customer behavior, product performance, sales trends, and customer segmentation**

With clean data, analysts write business-driven SQL queries to extract insights on customer lifetime value, popular products, peak shopping days, and inventory decisions.

Dataset Overview

The dataset contains customer information and transaction records simulating a real-world e-commerce platform.

Dataset 1: Customer Details (`customers` table)

Each row represents a unique customer with demographic information.

Dataset 2: Basket Details (`baskets` table)

Each row represents a transaction where a customer purchased a product.

<u>Data Volume</u>

Customers: 5,000+ records

Transactions: 18,000+ records

Unique Products: 3,000+

Project Workflow

1. Database & Table Creation

We start by creating SQL tables with appropriate data types:

```sql
-- Create customers table
CREATE TABLE customers (
    customer_id INTEGER,
    sex TEXT,
    customer_age INTEGER,
    tenure INTEGER
);

-- Create baskets table
CREATE TABLE baskets (
    customer_id INTEGER,
    product_id INTEGER,
    basket_date TEXT,
    basket_count INTEGER
);
```

2. Data Import

- Open DB Browser for SQLite

- Create a new database (project.db)
- **File → Import → Table from CSV file**
- Import Customer details.csv as table customers
- Import Basket details.csv as table baskets
- ✅ Check "Column names in first line"
- ✅ Check "Trim fields"
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3. Data exploration and cleaning

Check total records in each table

```
SELECT 'customers' as Table_Name, COUNT(*) as Total_Rows FROM customers
UNION ALL
SELECT 'baskets', COUNT(*) FROM baskets;
```

Preview Data Structure

```
SELECT * FROM customers LIMIT 5;
SELECT * FROM baskets LIMIT 5;
```

Check for Null values

```
SELECT
   COUNT(*) as total_rows,
   SUM(CASE WHEN customer_id IS NULL THEN 1 ELSE 0 END) as missing_customer_ids,
   SUM(CASE WHEN product_id IS NULL THEN 1 ELSE 0 END) as missing_product_ids
FROM baskets;
```

Fix Date Format

```
ALTER TABLE baskets ADD COLUMN purchase_date TEXT;

UPDATE baskets
SET purchase_date = SUBSTR(basket_date, 7, 4) || '-' ||
            SUBSTR(basket_date, 4, 2) || '-' ||
            SUBSTR(basket_date, 1, 2);
```

4. Business Insights

Customer Demographics

Gender breakdown of customers

Age group distribution

Most loyal customers (longest tenure)

Sales Performance

Overall business metrics

Best sales day ever